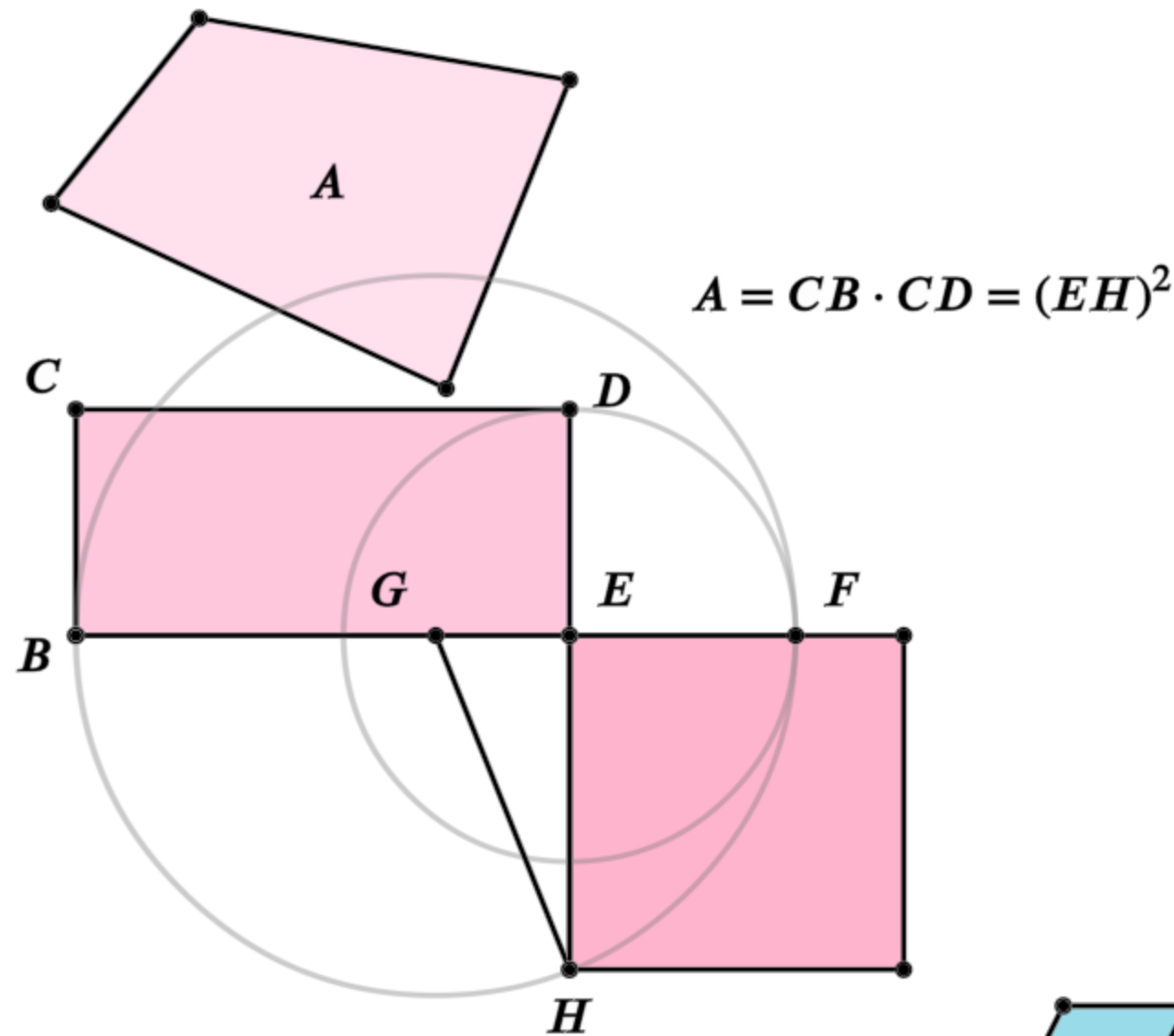


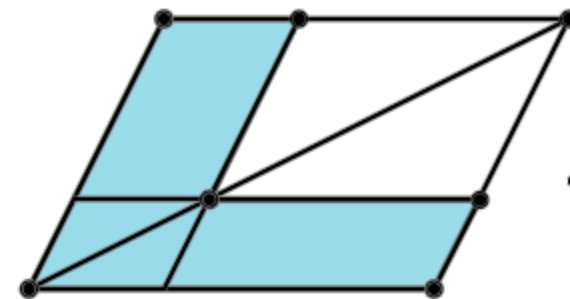
Euclid's Elements

Book II



It is a remarkable fact in the history of geometry, that the Elements of Euclid, written two thousand years ago, are still regarded by many as the best introduction to the mathematical sciences.

- Florian Cajori,
A History of Mathematics (1893)

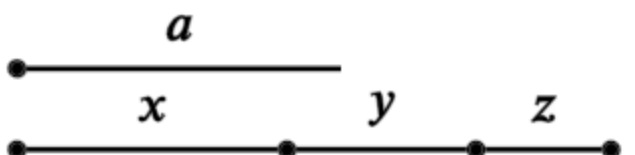


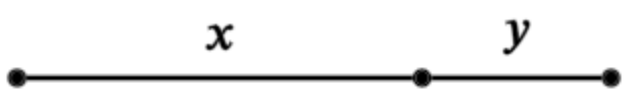
Definitions:

Any rectangular parallelogram is said to be contained by the two straight lines containing the right angle.

And in any parallelogrammic area let any one whatever of the parallelograms about its diameter with the two complements be called a gnomon.

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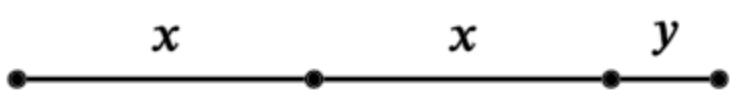
1. 
 $a(x + y + z) = ax + ay + az$

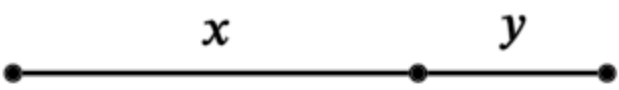
2. 
 $(x + y)^2 = x(x + y) + y(x + y)$

3. 
 $y(x + y) = yx + y^2$

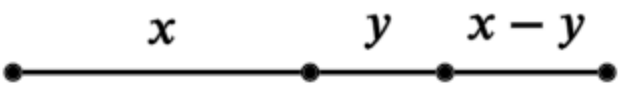
4. 
 $(x + y)^2 = x^2 + y^2 + 2xy$

5. 
 $(x + y)(x - y) + y^2 = x^2$

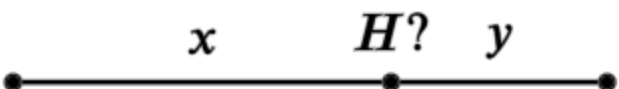
6. 
 $y(2x + y) + x^2 = (x + y)^2$

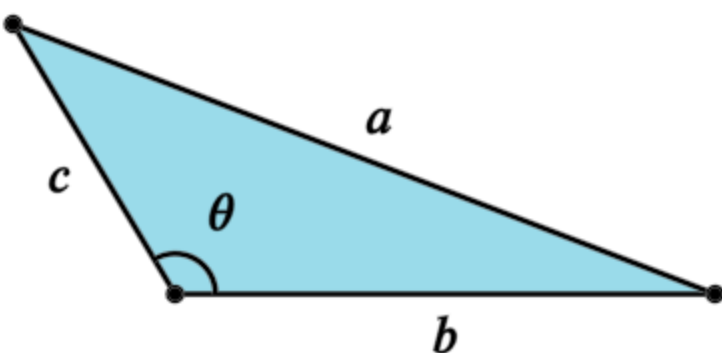
7. 
 $(x + y)^2 + y^2 = x^2 + 2y(x + y)$

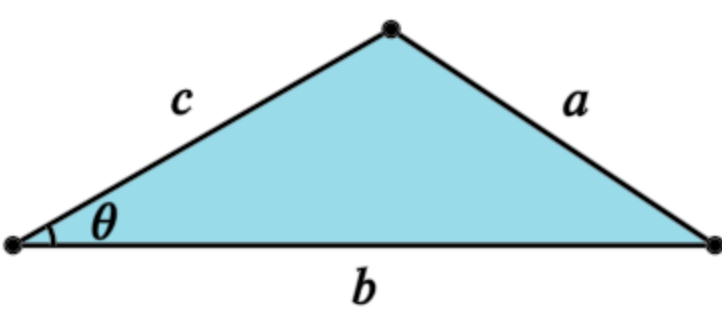
8. 
 $(x + 2y)^2 = 4y(x + y) + x^2$

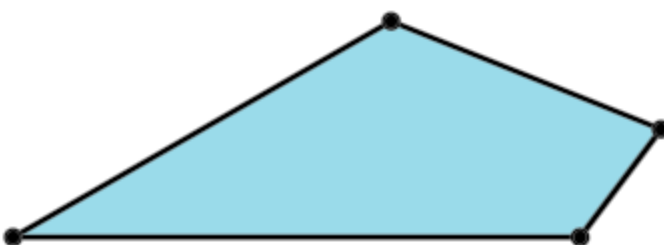
9. 
 $(x + y)^2 + (x - y)^2 = 2(x^2 + y^2)$

10. 
 $(2x + y)^2 + y^2 = 2(x^2 + (x + y)^2)$

11. 
Find H such that: $y(x + y) = x^2$

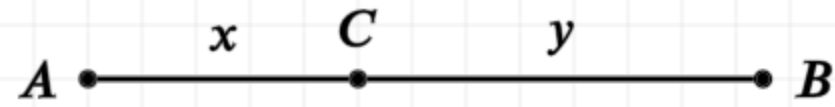
12. 
Cosine Law, $a^2 = b^2 + c^2 - 2bc \cos \theta$

13. 
Cosine Law. $a^2 = b^2 + c^2 - 2bc \cos \theta$

14. 
Construct a square equal to the polygon

Proposition 4 of Book 2

If a straight line be cut at random, the square on the whole is equal to the squares on the segments and twice the rectangle contained by the segments.



In other words

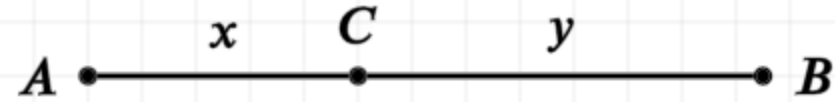
Let AB be a straight line, arbitrarily cut at point C

$$AB = AC + CB$$

$$a = x + y$$

Proposition 4 of Book 2

If a straight line be cut at random, the square on the whole is equal to the squares on the segments and twice the rectangle contained by the segments.



In other words

Let AB be a straight line, arbitrarily cut at point C

Then the square formed by line AB is equal in area to the sum of the squares formed by line CB and AC , plus twice the area of the rectangles formed by lines AC and CB

$$AB = AC + CB$$

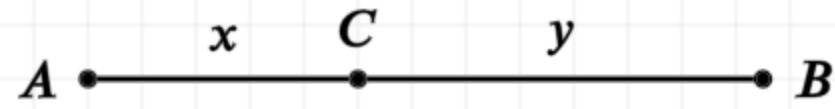
$$AB^2 = AC^2 + CB^2 + 2 \cdot AC \cdot CB$$

$$a = x + y$$

$$(x + y)^2 = x^2 + y^2 + 2xy$$

Proposition 4 of Book 2

If a straight line be cut at random, the square on the whole is equal to the squares on the segments and twice the rectangle contained by the segments.

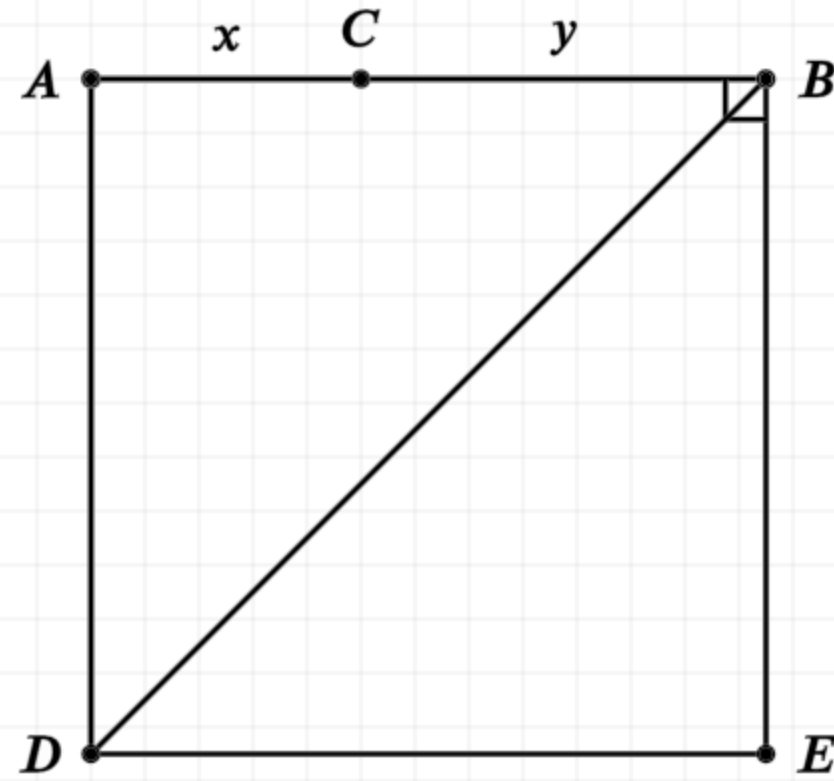


Construction:

$$AB = AC + CB$$

Proposition 4 of Book 2

If a straight line be cut at random, the square on the whole is equal to the squares on the segments and twice the rectangle contained by the segments.



$$AB = AD, \quad \angle ABE = \angle$$

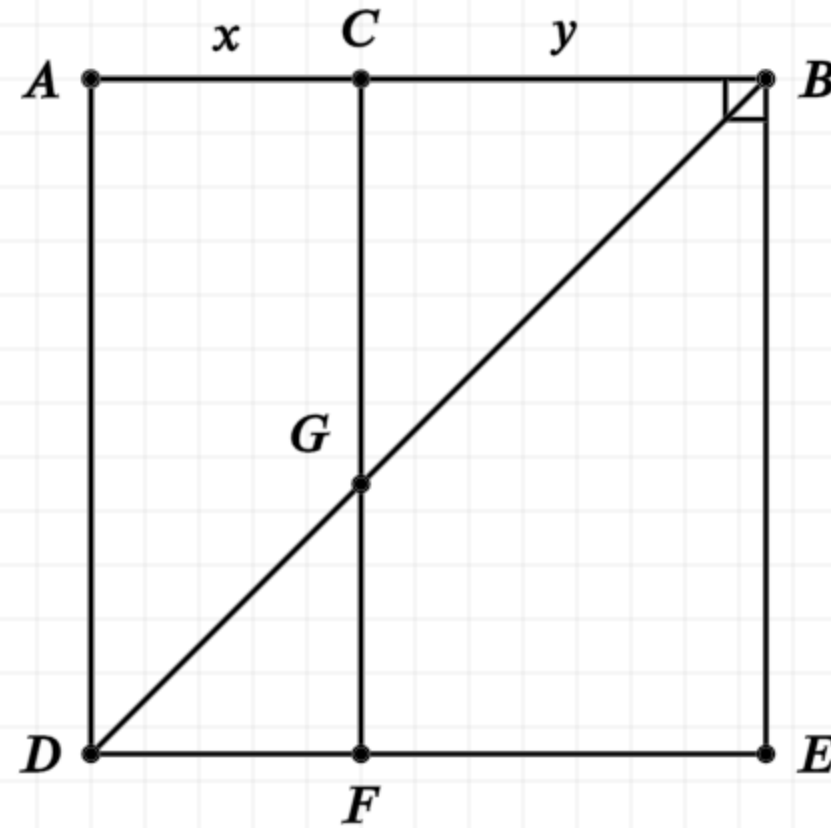
$$AB = AC + CB$$

Construction:

Draw a square ADEB on the line AB (I.46), and draw the diagonal BD

Proposition 4 of Book 2

If a straight line be cut at random, the square on the whole is equal to the squares on the segments and twice the rectangle contained by the segments.



$$AB = AD, \quad \angle ABE = \angle$$

$$AB = AC + CB$$

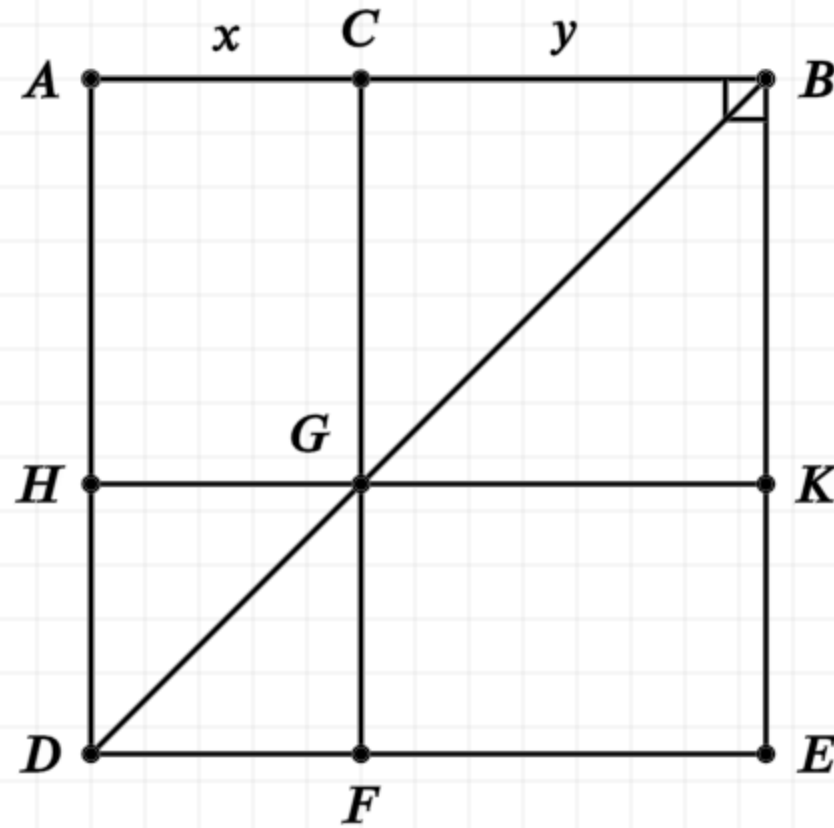
Construction:

Draw a square ADEB on the line AB (I.46), and draw the diagonal BD

Draw a line CF parallel to either AD or BE (I.31), labelling the intersection with the diagonal as G.

Proposition 4 of Book 2

If a straight line be cut at random, the square on the whole is equal to the squares on the segments and twice the rectangle contained by the segments.



$$AB = AD, \quad \angle ABE = \angle$$

$$AB = AC + CB$$

Construction:

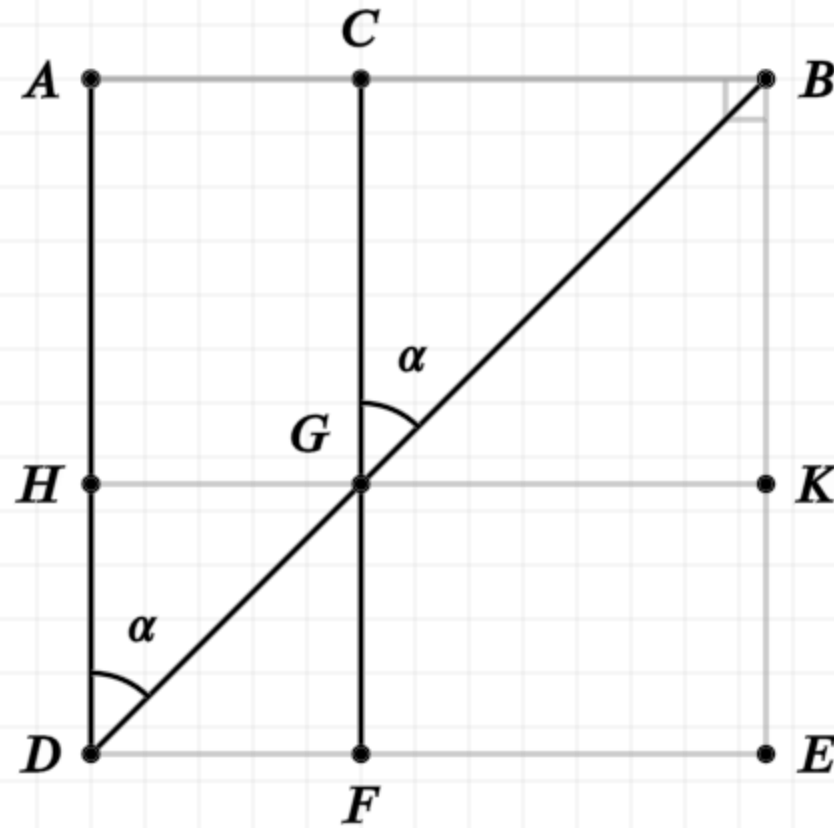
Draw a square ADEB on the line AB (I.46), and draw the diagonal BD

Draw a line CF parallel to either AD or BE (I.31), labelling the intersection with the diagonal as G.

Draw a line parallel to AB through the point G (I.31).

Proposition 4 of Book 2

If a straight line be cut at random, the square on the whole is equal to the squares on the segments and twice the rectangle contained by the segments.



$$AB = AD, \quad \angle ABE = \angle CGB = \alpha$$

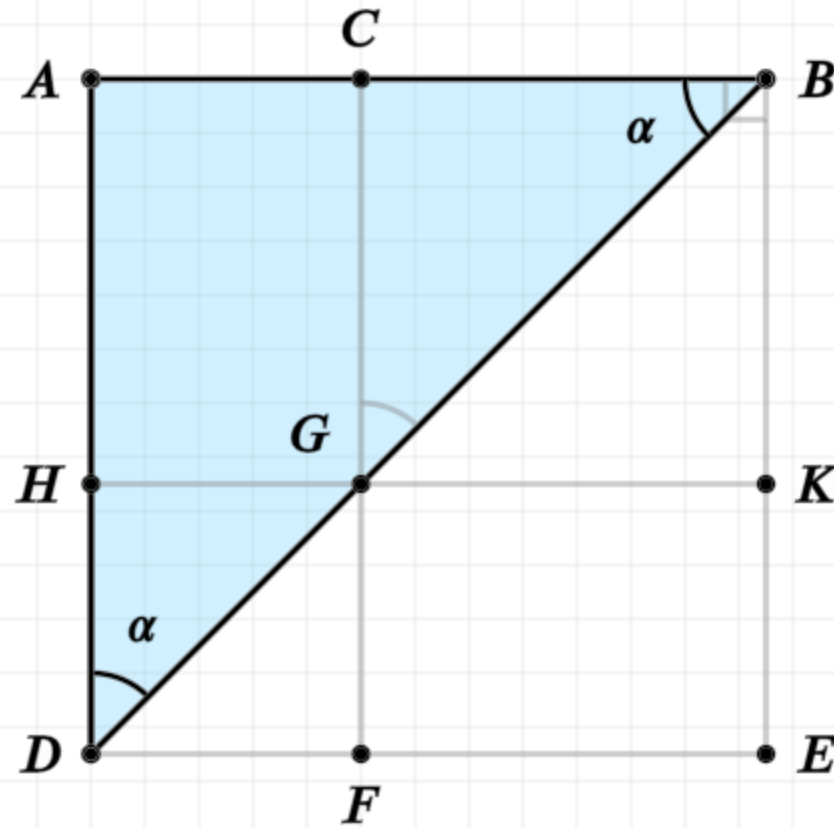
$$AB = AC + CB$$

Proof:

Since BD crosses two parallel lines, (AD and CF), then the exterior angle is equal to the interior and opposite angle (I.29)

Proposition 4 of Book 2

If a straight line be cut at random, the square on the whole is equal to the squares on the segments and twice the rectangle contained by the segments.



$$AB = AD, \quad \angle ABE = \angle$$

$$\angle CGB = \angle ADB$$

$$\angle ADB = \angle ABD$$

$$AB = AC + CB$$

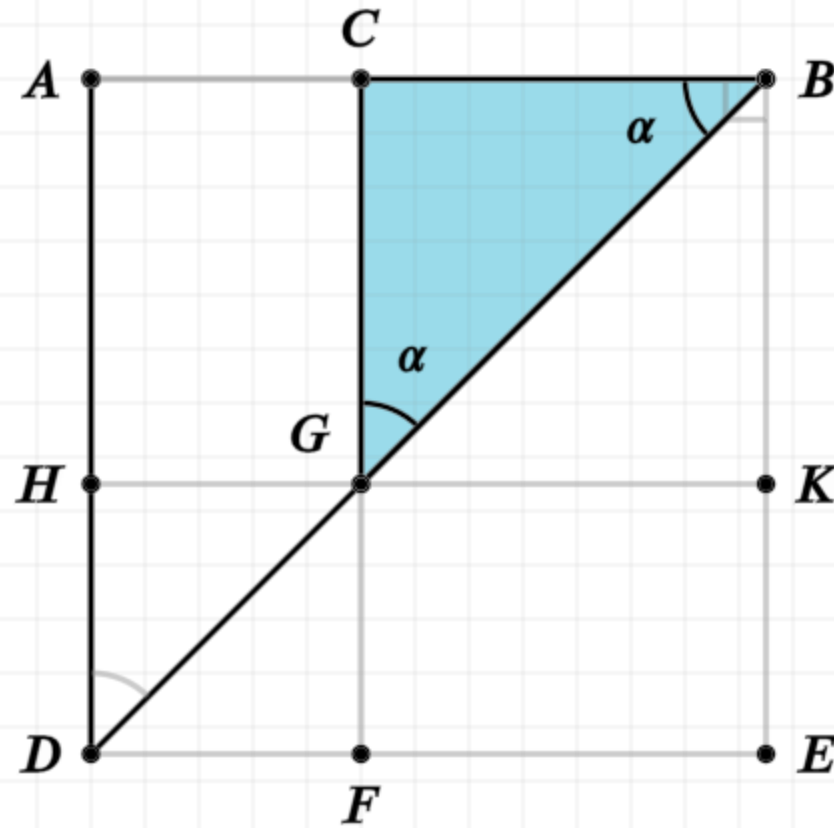
Proof:

Since BD crosses two parallel lines, (AD and CF), then the exterior angle is equal to the interior and opposite angle (I.29)

Since triangle ABD is an isosceles triangle, then the angles at the base are also equal (I.5)

Proposition 4 of Book 2

If a straight line be cut at random, the square on the whole is equal to the squares on the segments and twice the rectangle contained by the segments.



$$AB = AD, \quad \angle ABE = \angle$$

$$\angle CGB = \angle ADB$$

$$\angle ADB = \angle ABD$$

$$CG = CB$$

$$AB = AC + CB$$

Proof:

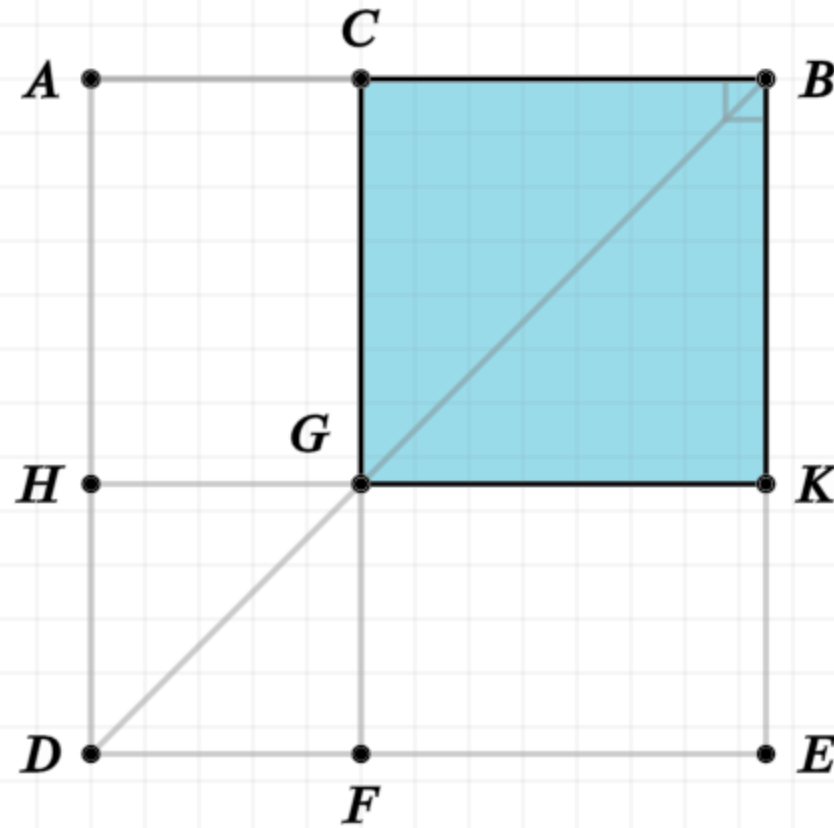
Since BD crosses two parallel lines, (AD and CF), then the exterior angle is equal to the interior and opposite angle (I.29)

Since triangle ABD is an isosceles triangle, then the angles at the base are also equal (I.5)

Since triangle BCG has two equal angles, it is an isosceles triangle (I.6), therefore the sides of the triangle are equal

Proposition 4 of Book 2

If a straight line be cut at random, the square on the whole is equal to the squares on the segments and twice the rectangle contained by the segments.



$$AB = AC + CB$$

$$AB = AD, \quad \angle ABE = \angle$$

$$\angle CGB = \angle ADB$$

$$\angle ADB = \angle ABD$$

$$CG = CB$$

$$CB \parallel GK, \quad CG \parallel BK$$

$$\therefore CB = GK, \quad CG = BK$$

Proof:

Since BD crosses two parallel lines, (AD and CF), then the exterior angle is equal to the interior and opposite angle (I.29)

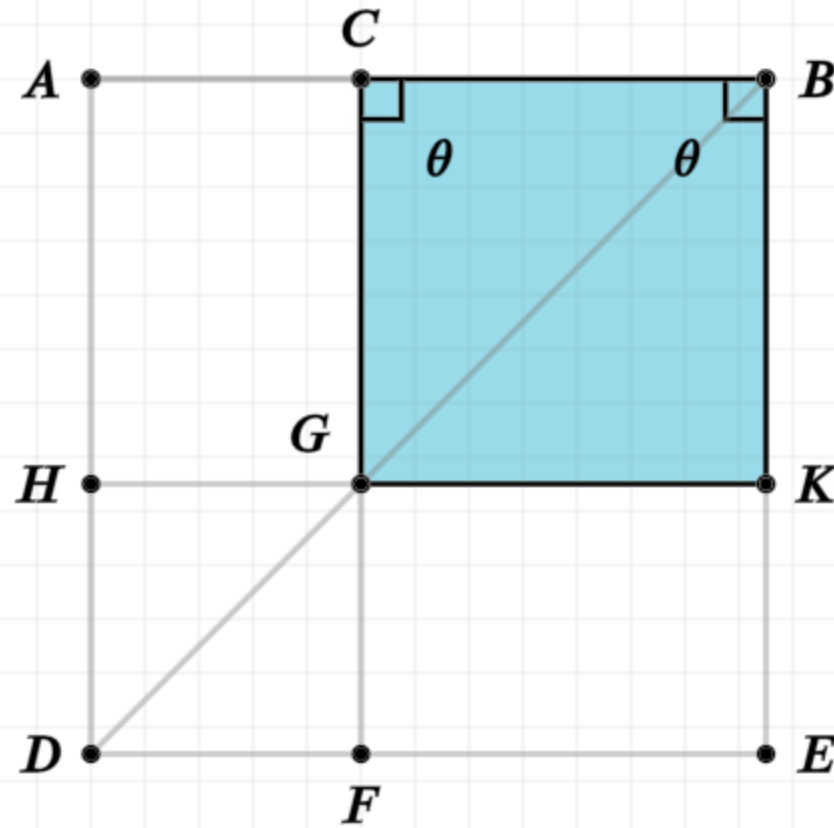
Since triangle ABD is an isosceles triangle, then the angles at the base are also equal (I.5)

Since triangle BCG has two equal angles, it is an isosceles triangle (I.6), therefore the sides of the triangle are equal

CB and GK are parallel, CG and BK are parallel, so the opposite sides equal on another (I.34), and since CB equals CG, all sides are equal, therefore **CK is equilateral**

Proposition 4 of Book 2

If a straight line be cut at random, the square on the whole is equal to the squares on the segments and twice the rectangle contained by the segments.



$$AB = AC + CB$$

$$AB = AD, \quad \angle ABE = L$$

$$\angle CGB = \angle ADB$$

$$\angle ADB = \angle ABD$$

$$CG = CB$$

$$CB \parallel GK, \quad CG \parallel BK$$

$$\therefore CB = GK, \quad CG = BK$$

$$\angle GCB + \angle ABE = L + L$$

$$\angle GCB = L$$

Proof:

Since BD crosses two parallel lines, (AD and CF), then the exterior angle is equal to the interior and opposite angle (I.29)

Since triangle ABD is an isosceles triangle, then the angles at the base are also equal (I.5)

Since triangle BCG has two equal angles, it is an isosceles triangle (I.6), therefore the sides of the triangle are equal

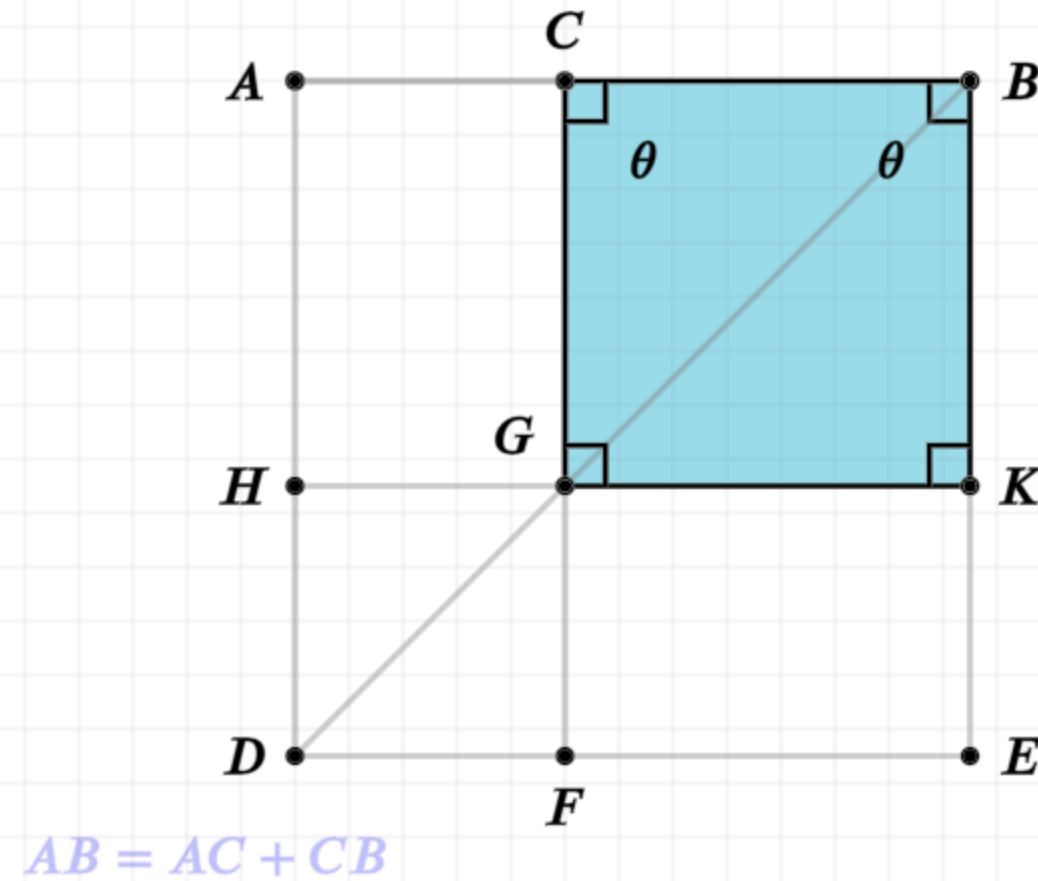
CB and GK are parallel, CG and BK are parallel, so the opposite sides equal on another (I.34), and since CB equals CG, all sides are equal, therefore **CK is equilateral**

Since CG is parallel to BK, then the sum of the interior angles is two right angles (I.29)

We know that angle CBK is right, so then angle BCG is also right.

Proposition 4 of Book 2

If a straight line be cut at random, the square on the whole is equal to the squares on the segments and twice the rectangle contained by the segments.



$$AB = AD, \quad \angle ABE = L$$

$$\angle CGB = \angle ADB$$

$$\angle ADB = \angle ABD$$

$$CG = CB$$

$$CB \parallel GK, \quad CG \parallel BK$$

$$\therefore CB = GK, \quad CG = BK$$

$$\angle GCB + \angle ABE = L + L$$

$$\angle GCB = L$$

$$\angle CGK = L$$

$$\angle BKG = L$$

Proof:

Since BD crosses two parallel lines, (AD and CF), then the exterior angle is equal to the interior and opposite angle (I.29)

Since triangle ABD is an isosceles triangle, then the angles at the base are also equal (I.5)

Since triangle BCG has two equal angles, it is an isosceles triangle (I.6), therefore the sides of the triangle are equal

CB and GK are parallel, CG and BK are parallel, so the opposite sides equal on another (I.34), and since CB equals CG, all sides are equal, therefore **CK is equilateral**

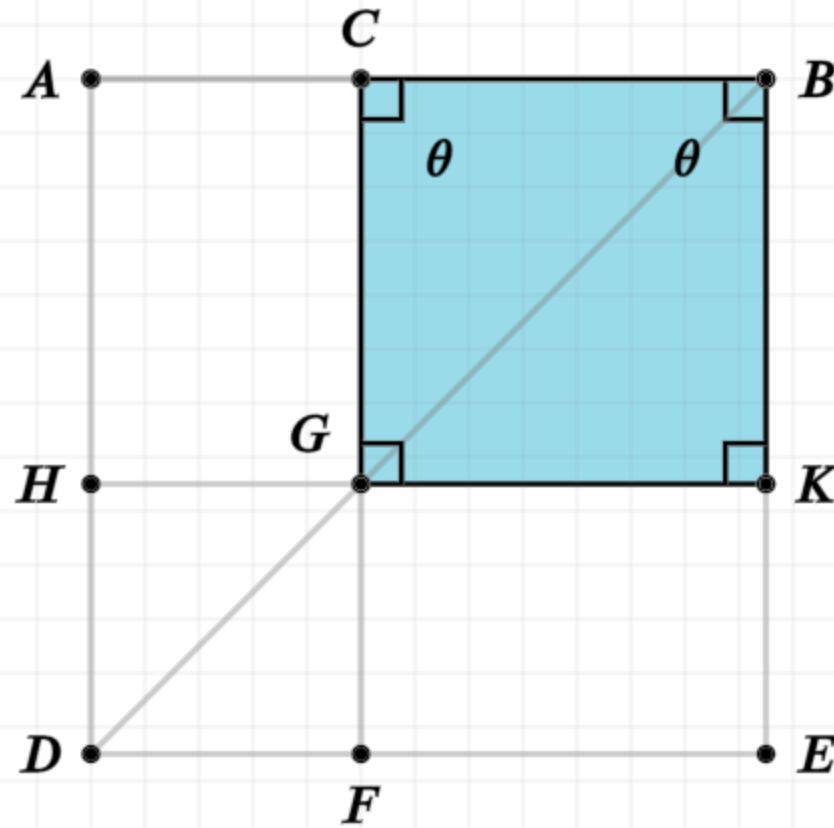
Since CG is parallel to BK, then the sum of the interior angles is two right angles (I.29)

We know that angle CBK is right, so then angle BCG is also right.

The angles opposite one another are equal in a parallelogram, so the other two angles are right angles as well (I.34).

Proposition 4 of Book 2

If a straight line be cut at random, the square on the whole is equal to the squares on the segments and twice the rectangle contained by the segments.



$$AB = AC + CB$$

$$AB = AD, \quad \angle ABE = \angle$$

$$\angle CGB = \angle ADB$$

$$\angle ADB = \angle ABD$$

$$CG = CB$$

$$CB \parallel GK, \quad CG \parallel BK$$

$$\therefore CB = GK, \quad CG = BK$$

$$\angle GCB + \angle ABE = L + L$$

$$\Delta GCB = L$$

$$\triangle CGK = L$$

$\angle BKG = \angle L$

■ $CK = CB^2$

Proof:

Since BD crosses two parallel lines, (AD and CF), then the exterior angle is equal to the interior and opposite angle (I.29)

Since triangle ABD is an isosceles triangle, then the angles at the base are also equal (1.5)

Since triangle BCG has two equal angles, it is an isosceles triangle (1.6), therefore the sides of the triangle are equal

CB and GK are parallel, CG and BK are parallel, so the opposite sides equal on another (I.34), and since CB equals CG, all sides are equal, therefore **CK is equilateral**

Since CG is parallel to BK, then the sum of the interior angles is two right angles (I.29)

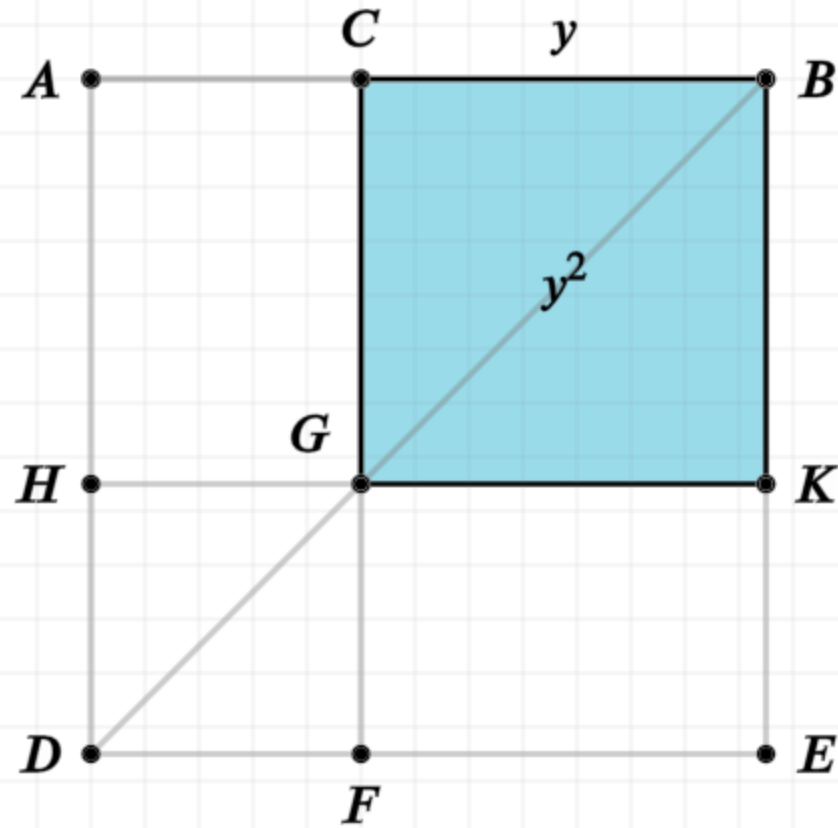
We know that angle CBK is right, so then angle BCG is also right.

The angles opposite one another are equal in a parallelogram, so the other two angles are right angles as well (I.34).

So CK is a square, equal to the square of CB

Proposition 4 of Book 2

If a straight line be cut at random, the square on the whole is equal to the squares on the segments and twice the rectangle contained by the segments.



$$AB = AC + CB$$

$$CG = CB$$

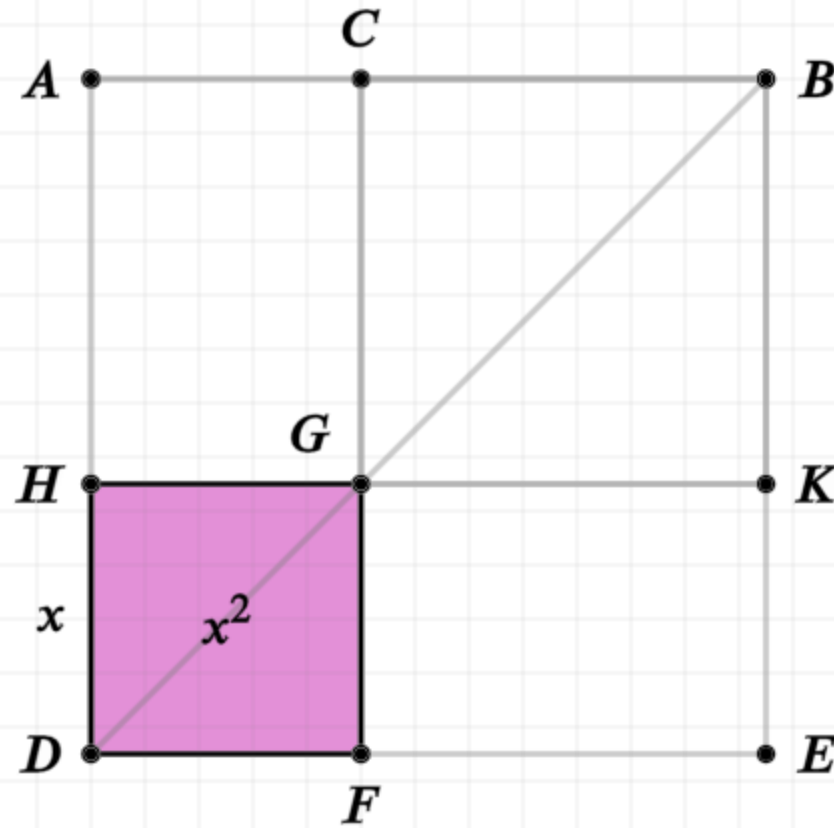
$$\text{■ } CK = CB^2$$

Proof:

CK is a square, equal to the square of CB

Proposition 4 of Book 2

If a straight line be cut at random, the square on the whole is equal to the squares on the segments and twice the rectangle contained by the segments.



$$AB = AC + CB$$

$$CG = CB$$

$$\text{CK} = CB^2$$

$$\text{HF} = AC^2$$

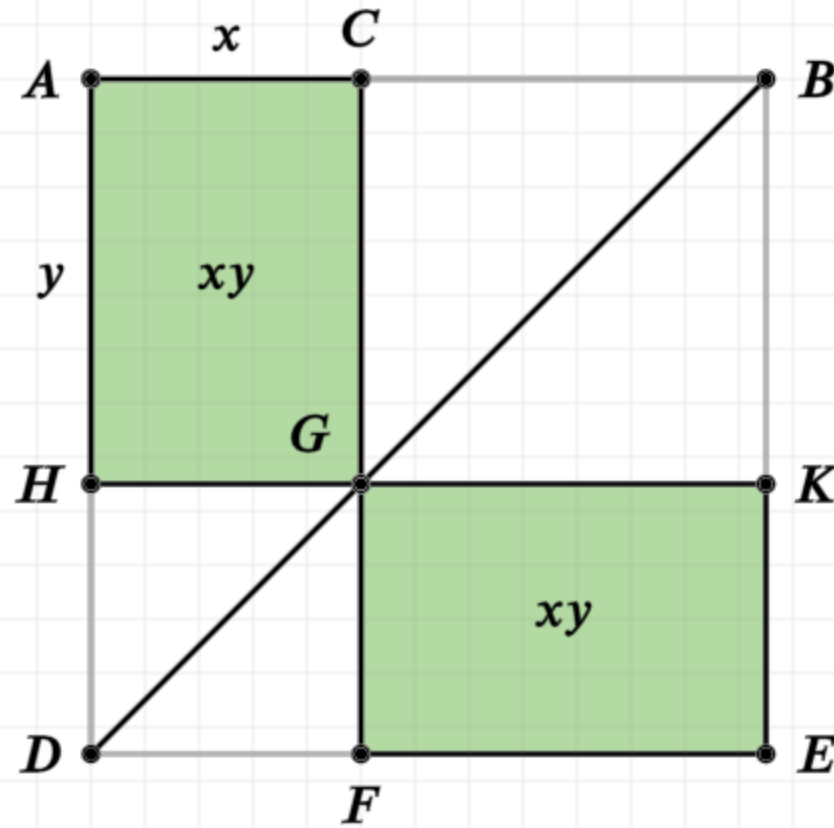
Proof:

CK is a square, equal to the square of CB

Similarly, HCFG is a square, equal to the square of AC

Proposition 4 of Book 2

If a straight line be cut at random, the square on the whole is equal to the squares on the segments and twice the rectangle contained by the segments.



$$AB = AC + CB$$

$$CG = CB$$

$$CK = CB^2$$

$$HF = AC^2$$

$$AG = GE$$

$$AG = AC \cdot CG = AC \cdot CB$$

Proof:

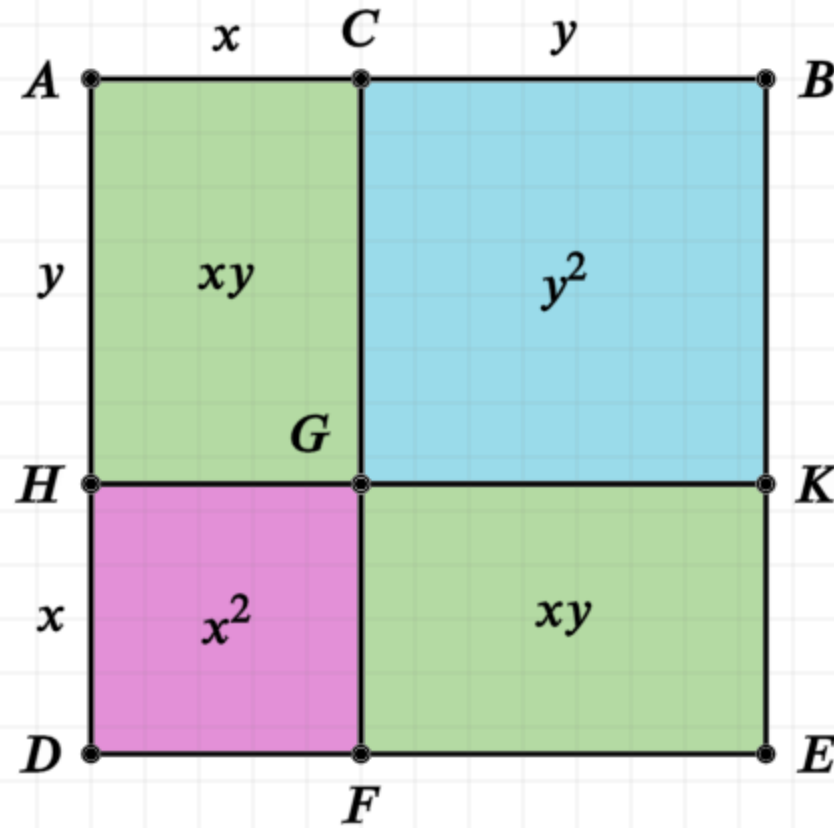
CK is a square, equal to the square of CB

Similarly, HFG is a square, equal to the square of AC

Rectangles AG and GE are equal (complements of a parallelogram) (I.43), and are equal to the rectangle formed from lines AC and CB

Proposition 4 of Book 2

If a straight line be cut at random, the square on the whole is equal to the squares on the segments and twice the rectangle contained by the segments.



$$AB = AC + CB$$

$$CG = CB$$

$$\text{CK} = CB^2$$

$$\text{HF} = AC^2$$

$$\text{AG} = \text{GE}$$

$$\text{AG} = AC \cdot CG = AC \cdot CB$$

$$\square AE = \text{CK} + \text{HF} + \text{AG} + \text{GE}$$

$$AB^2 = CB^2 + AC^2 + 2(AC \cdot CB)$$

Proof:

CK is a square, equal to the square of CB

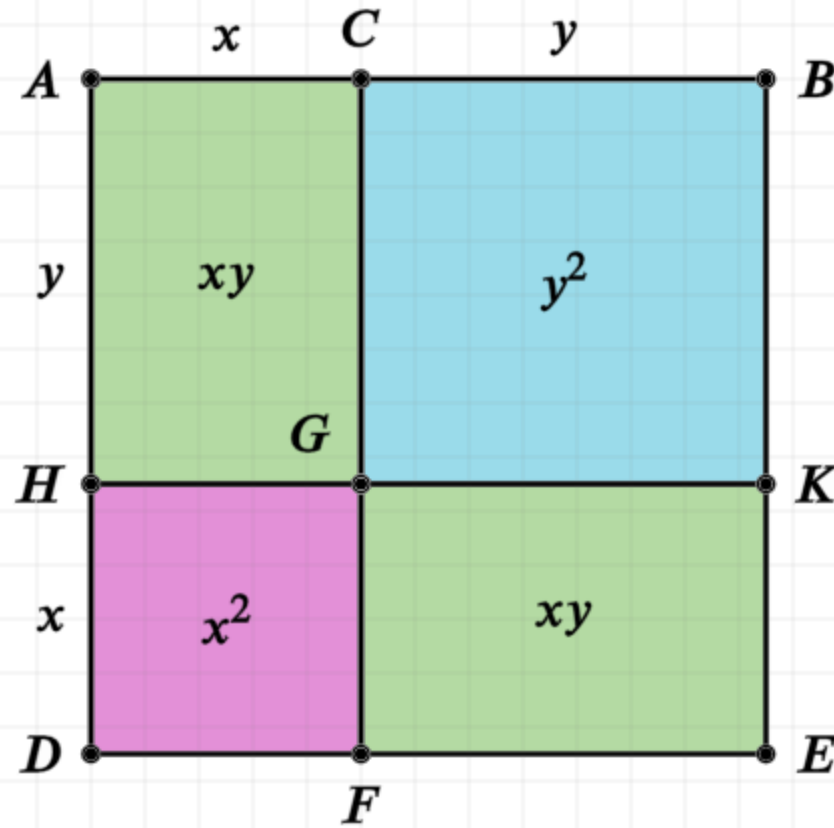
Similarly, HCFG is a square, equal to the square of AC

Rectangles AG and GE are equal (complements of a parallelogram) (I.43), and are equal to the rectangle formed from lines AC and CB

The sum of all the rectangles equals the square on AB

Proposition 4 of Book 2

If a straight line be cut at random, the square on the whole is equal to the squares on the segments and twice the rectangle contained by the segments.



$$AB = AC + CB$$

$$CG = CB$$

$$\text{CK} = CB^2$$

$$\text{HF} = AC^2$$

$$\text{AG} = \text{GE}$$

$$\text{AG} = AC \cdot CG = AC \cdot CB$$

$$\square AE = \text{CK} + \text{HF} + \text{AG} + \text{GE}$$

$$AB^2 = CB^2 + AC^2 + 2(AC \cdot CB), \quad (x + y)^2 = x^2 + y^2 + 2xy$$

Proof:

CK is a square, equal to the square of CB

Similarly, HCFG is a square, equal to the square of AC

Rectangles AG and GE are equal (complements of a parallelogram) (I.43), and are equal to the rectangle formed from lines AC and CB

The sum of all the rectangles equals the square on AB

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